

TD 7 - Apprentissage non supervisé

Exercice 1 – Spectral Clustering

Consider an undirected graph $G = (V, E)$ with n vertices, $V = \{v_1, \dots, v_n\}$. Associate with it a non-negative weight matrix $W = (w_{ij})_{i,j=1,\dots,n}$ where $w_{ij} \geq 0$ (symmetric). The nodes of this graph represent the examples to be clustered, and the weights represent the similarities between examples : the stronger the weight between two nodes, the more similar the associated examples are. One of the advantages of spectral clustering is that it operates on a concept of similarity (like kernels in SVMs) rather than solely geometrically on an Euclidean description of our examples. Of course, any Euclidean description can be represented by a similarity (e.g., by taking the exponential of the inverse of the distance). The principle of spectral clustering is to form partitions such that the nodes within the same partition have very strong connectivity (the weights within each cluster must be very high) and the inter-cluster weights are as low as possible—that is, there should be weak connectivity between the vertices of two disjoint clusters. In graph theory terms, it is about partitioning the graph such that the associated *cut* eliminates the connections with the weakest weights.

The quality measure of a cut in k partitions $A_1, \dots, A_k$ is defined by :

$$Rcut(A_1, \dots, A_k) = \frac{1}{2} \sum_{i=1}^k \frac{W(A_i, \bar{A}_i)}{|A_i|}$$

where $W(A, B) = \sum_{i \in A, j \in B} w_{ij}$, $\bar{A}$ is the complement of A in V , and $|A|$ is the number of vertices in A . Thus, spectral clustering seeks to minimize :

$$\min_{A_1, \dots, A_k} Rcut(A_1, \dots, A_k)$$

with $A_1, \dots, A_k$ being a partition of the vertices of V .

The algorithm is based on the concept of the Laplacian of a graph : let W be the weight matrix, and $D = (d_i)_{i=1\dots n}$ be the diagonal matrix such that $d_i = \sum_{j=1}^n w_{ij}$, the Laplacian of the graph is the matrix $L = D - W$.

The goal of this exercise is to examine an interpretation of this in terms of graph partitioning.

Reminder of Linear Algebra : A matrix $M \in \mathbb{R}^{d \times d}$ is symmetric and positive semi-definite if it satisfies one of the following equivalent properties :

- $\forall \mathbf{x} \in \mathbb{R}^d, \mathbf{x}^T M \mathbf{x} \geq 0$
- All eigenvalues are non-negative

A vector $\mathbf{v} \in \mathbb{R}^d$ and a scalar $\lambda \in \mathbb{R}$ are said to be an eigenvector and eigenvalue if $M\mathbf{v} = \lambda\mathbf{v}$. Every real symmetric matrix A has d independent eigenvectors. Therefore, A is diagonalizable : $A = Q^T \Lambda Q$ with Q an orthogonal matrix composed of the orthonormal eigenvectors ($\|q_i\|^2 = 1$ and $\forall i, j, q_i \cdot q_j = 0$), and Λ is diagonal with the eigenvalues corresponding to the eigenvectors.

Q 1.1 We will first prove some properties of the Laplacian that we will use later in the exercise. Show that :

Q 1.1.1 $\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x}^T L \mathbf{x} = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - x_j)^2$

Q 1.1.2 L is symmetric and positive semi-definite.

Q 1.1.3 The smallest eigenvalue of L is 0, and the associated eigenvector is the vector $\mathbf{e} = (1, \dots, 1)^T$ of size n .

Q 1.2 Consider the case of separation into 2 clusters, A and $\bar{A}$, with $k = 2$. The goal is to transform the expression $\min_{A \subset V} Rcut(A, \bar{A})$ into a continuous optimization problem.

Let $\mathbf{x} \in \mathbb{R}^n$ such that $x_i = \sqrt{\frac{|\bar{A}|}{|A|}}$ if $x_i \in A$, and $x_i = -\sqrt{\frac{|A|}{|\bar{A}|}}$ if $x_i \in \bar{A}$. The value of x_i thus indicates whether node i belongs to the first or second cluster. The vector $\mathbf{x}$ encodes all possible partitionings into two clusters.

Show that :

Q 1.2.1 $\mathbf{x}^T L \mathbf{x} = n \cdot Rcut(A, \bar{A})$

Q 1.2.2 $\mathbf{x}^T \mathbf{e} = \sum_{i=1}^n x_i = 0$

Q 1.2.3 $\|\mathbf{x}\|^2 = \mathbf{x}^T \mathbf{x} = n$. How do you interpret this constraint and the one from the previous question ?

We have just shown that the problem $\min_{A \subset V} Rcut(A, \bar{A})$ is equivalent to $\min_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^T L \mathbf{x}$ under the constraint of a vector $\mathbf{x}$ encoding the partitioning : $\mathbf{x} = (x_1, \dots, x_N)$, with $x_i = \alpha$ or $x_i = \beta$, where $\alpha = \sqrt{\frac{|\bar{A}|}{|A|}}$ and $\beta = -\sqrt{\frac{|A|}{|\bar{A}|}}$. The constants α and β can be arbitrary, but if chosen well, they allow for balancing the partitioning and obtaining exactly the $Rcut$ cost measure.

This problem is NP-hard : it involves choosing combinatorially the correct assignment of each node into one of the two partitions. To relax it and be able to address it through continuous optimization, we will remove the constraint on the vector $\mathbf{x}$ while introducing the two constraints $\mathbf{x}^T \mathbf{x} = n$ and $\mathbf{x}^T \mathbf{e} = 0$ to ensure solutions that are not too far from our discrete optimum.

Q 1.3 Consider the optimization problem $\min_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^T L \mathbf{x}$ under the constraint $\mathbf{x}^T \mathbf{x} = n$. Show that the solution to this problem is an eigenvector $e^{(i)}$ of L , and that for this vector, $e^{(i)T} L e^{(i)} = n \lambda_i$ where λ_i is the eigenvalue associated with $e^{(i)}$ (use $\nabla_{\mathbf{x}} \mathbf{x}^T \mathbf{x} = 2\mathbf{x}$ and $\nabla_{\mathbf{x}} \mathbf{x}^T A \mathbf{x} = (A + A^T)\mathbf{x}$). Deduce the solution to the problem.

Q 1.4 Consider the optimization problem $\min_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^T L \mathbf{x}$ under the constraints $\mathbf{x}^T \mathbf{x} = n$ and $\mathbf{x}^T \mathbf{e} = 0$. Using the previous question, give (without calculations) the solution to this problem. Let e' denote this vector.

Q 1.5 How can we use the eigenvector e' to construct a solution to the clustering problem for $k = 2$? For reference, the general algorithm is as follows, with k being the desired number of clusters. The proof is almost the same as for the $k = 2$ case, but with as many indicator vectors for cluster membership as there are clusters :

- Calculate the Laplacian L
- Compute the first k eigenvectors of L
- Let $U \in \mathbb{R}^{n \times k}$ be the matrix containing the k eigenvectors $u_1, \dots, u_k$ in columns
- For $i = 1 \dots n$, let $y_i \in \mathbb{R}^k$ be the corresponding i -th row of U
- Perform clustering of the $(y_i)_{i=1 \dots n}$ in $\mathbb{R}^k$ using a k -means algorithm. Denote the obtained clusters as $C_1, \dots, C_k$.
- The final clusters are $A_1, \dots, A_k$ with $A_i = \{v_j \in V \mid y_j \in C_i\}$

For more details, you can refer to this article : [this paper](#).